wheelset-s2pinn: Phase 0 report (gate G0)

Generated from **results/** by experiments/make_g0_report.py. Gate file commit: 31c07e59cefbce51de80a0641ea834e9fcdf40d5; generated 2026-10-11T22:31:41Z. Every number below is read from a results JSON written by an experiment script.

1. Gate summary

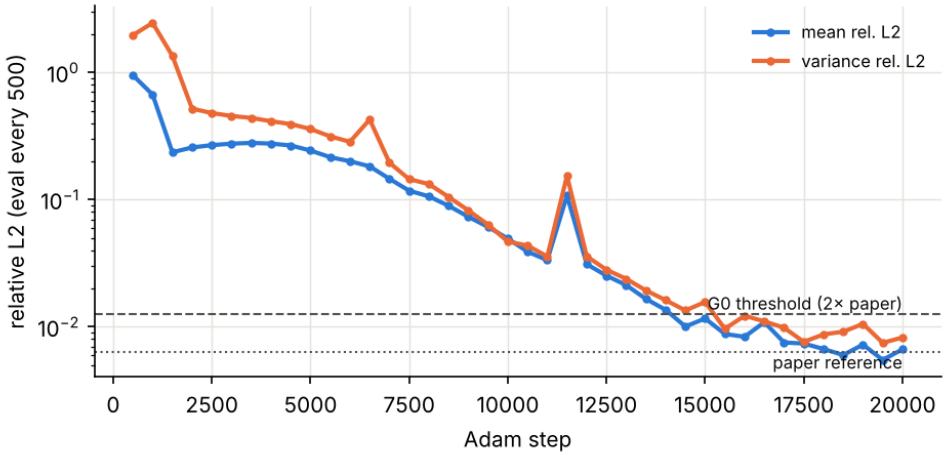
Item	Status	Key numbers
(a) S²-PINN reproduces stochastic diffusion within 2× of the paper's mean error (Nc=4096 vs Table 14)	PASS	mean rel. L2 = 6.709e-03 vs threshold 0.0127 (1.06× paper)
(b) SDOF shows a clear Π² trend in the required gPC degree	PASS AS REFRAMED (accepted 2026-10-11): degree ~ Pi for uniform, ~ Pi^2 for Gaussian inputs	Legendre: p ∝ $\Pi^{0.91\pm0.01}$ (R^2=0.994); Hermite: overall $\Pi^{1.59}$, local slope → 1.93 at large Π
(c) Eq. (41) within 10% of measured effective damping, or measured relation reported	MET VIA FALLBACK: Eq. 41 misses 10%; measured relation reported; correction dropped (agreed 2026-10-11)	transient/sdof: median err 0.248; transient/2dof: median err 0.356; forced/sdof: median err 0.831; forced/2dof: median err 3.26
(d) Literature stop rule (method already published?)	NOT TRIGGERED	see docs/lit_review.md

2. (a) S²-PINN stochastic-diffusion reproduction

Quantity	This run (seed 42, Nc=4096)	Paper, Table 14 (Nc=4096) (5 seeds)
Mean rel. L2	6.709e-03	6.340e-03
Variance rel. L2	8.784e-03	0.013
Calibration error	0.0188	0.0185
G0 threshold (2× paper mean)	0.0127	
Device / budget	cpu (x86_64), 20k steps	T4, 20k steps
Wall time	5.2 h	≈90 min at Nc=32768 (Table 18)

Paper values in the right-hand column are quoted from the S²-PINN paper (Tables 1–3, 18) for comparison. Runner: resumable replay of spinn.train.train loop; s2pinn commit 1cd6fa2. CPU run: wall time and memory are not comparable with the paper's T4 figures

Deviation (approved): Nc=4096 instead of the paper default 32768 (PDE batch only, as the released --colloc flag does); compared with the paper's Table 14 row for this Nc. Table 14's own Nc=32768 row (6.43e-3) differs from Table 1 (3.71e-3); the paper does not explain the gap.



CPU cost on this container (4 cores, no GPU) at the paper default Nc=32768: 4.09 s per step, i.e. 22.7 h for 20k steps (hence the Nc=4096 run, whose 5.2 h includes periods sharing the CPU with other jobs). The 1 GB / 90 min cost limit refers to a T4 and cannot be checked on this hardware. This run is a single seed (42); the paper reports the mean of 5 seeds.

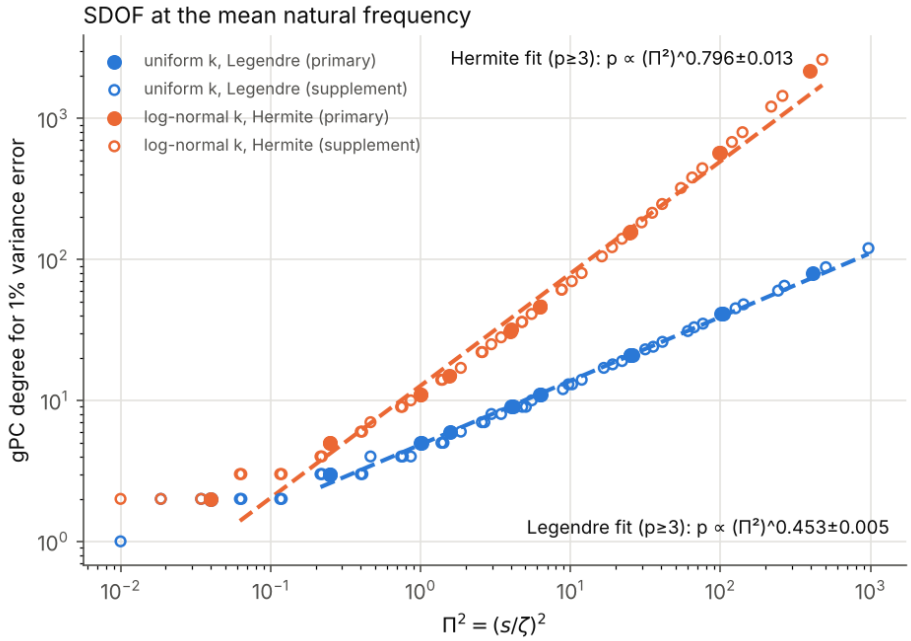
Runner checks: 20-step replay mean rel. L2 = 1.0082898 vs the released train() 1.0082900. Uninterrupted vs stopped-and-resumed 30-step runs: 0.9955707788 vs 0.9955707788.

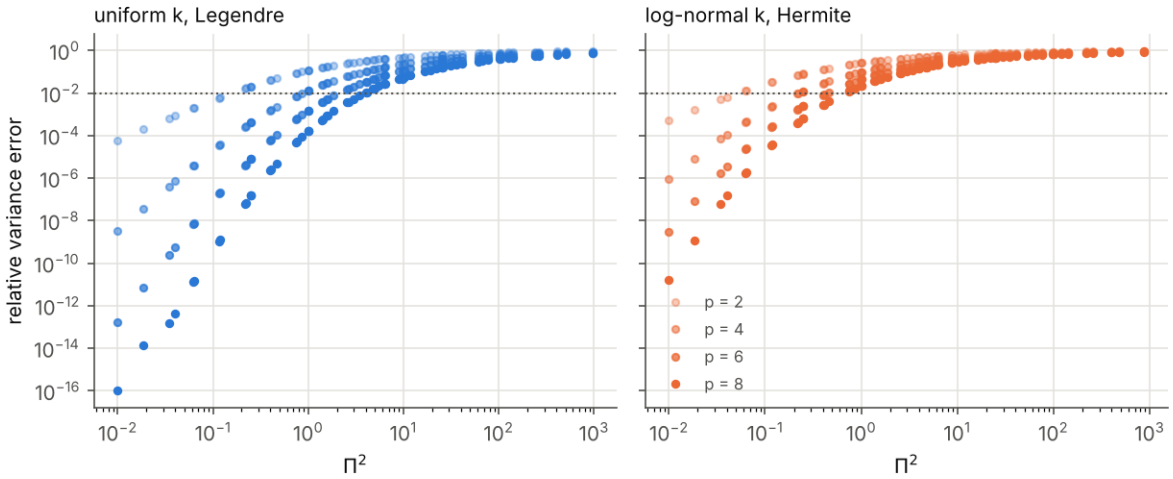
3. (b) H1 on an SDOF oscillator

Receptance of x'' + c x' + k x = $e^{i\omega t}$ with random stiffness (CoV ∈ 2%, 5%, 10%, 20%) and deterministic damper (ζ ∈ 0.5%, 1%, 2%, 5%), plus a 12-point CoV supplement. Required degree = smallest p with relative variance error ≤ 1% (orthonormal projection by quadrature). Π = s/ζ with s the natural-frequency CoV.

Family (input law)	slope of log p vs log Π (p≥3)	R^2	n	local slope by Π bin	p/Π or p/Π² by bin
Legendre (uniform k)	0.905 ± 0.010	0.9939	56	[0.5,2): 0.93 [2,5): 1.02 [5,10): 0.94 [10,40): 0.95	p/Π [0.5,2): 4.69 [2,5): 4.16 [5,10): 4.06 [10,40): 3.98
Hermite (log-normal k)	1.592 ± 0.025	0.9859	59	[0.5,2): 1.41 [2,5): 1.77 [5,10): 1.87 [10,40): 1.93	p/Π² [0.5,2): 10.09 [2,5): 6.92 [5,10): 5.89 [10,40): 5.57

Collapse onto Π: pairs of primary-grid cases with Π within 2% but different ζ and CoV need 9/9, 21/21, 21/21, 41/41, 5/5, 11/11, 21/21, 3/3, 32/31, 157/156, 157/154, 575/567, 11/11, 46/46, 156/154, 5/5 (required degrees).





Family	leading-order prediction	Π bin: measured p/Π (or p/Π²), measured/predicted
Legendre (uniform)	p ≈ (√3/2) Π ln(1/ε) = 3.99 Π	[0.5,2): 4.69, ratio 1.09 [2,5): 4.16, ratio 1.05 [5,10): 4.06, ratio 1.04 [10,40): 3.98, ratio 1.01

Family	leading-order prediction	Π bin: measured p/Π (or p/Π²), measured/predicted
Hermite (log-normal)	$p \approx \Pi^2 \ln^2(1/\varepsilon)/4 = 5.30\,\Pi^2$	[0.5,2): 10.09, ratio 1.90 [2,5): 6.92, ratio 1.30 [5,10): 5.89, ratio 1.12 [10,40): 5.57, ratio 1.06

Derivation: docs/derivations/h1_degree.md (Bernstein ellipse for Legendre; strip analyticity with Hille's rate for Hermite). Reworded G2, $\Pi \le 1.5$ with uniform inputs, needs $p = 6$ measured (leading order 5.98) for $\varepsilon = 0.01$ relative variance error.

Reading. The trend is clear, but the exponent depends on the input law. For bounded (uniform/Legendre) inputs, which is how the paper treats θ and W, the required degree grows like Π, not Π². Π² appears only asymptotically for Gaussian (Hermite) inputs. This matches approximation theory (Bernstein ellipse vs strip of analyticity) and the root-spreading result of Jacquelin et al. (2016).

4. (c) Eq. (41): residual-induced damping bias

Derivation (docs/derivations/eq41.md): with a long-window residual loss, an enforced initial condition and a relative unresolved residual floor R, minimising over the decay rate gives $\lambda^{*2} = \lambda_0^2 + (R/2\omega)^2$ with $\lambda_0 = \zeta\omega$. This is Eq. (41) once ζ is read as a decay rate. The steady-state counterpart is ridge shrinkage $X^* = (A^H A + R^2 I)^{-1} A^H F$, a different law.

Setting	n	median $\lvert\Delta\rvert/\lambda$ (Eq. 41)	within 10%	median bias / λ_0	alt. prediction (median err)	measured $\lambda^2-\lambda_0^2 = c\,(R/2\omega)^\beta$
transient/sdof	60	0.248	32%	0.973	window: 0.347	c=4.07, β=2.40±0.08 (n=58)
transient/2dof	120	0.356	6%	4.84	window: 0.396	c=2.72, β=1.88±0.07 (n=120)
forced/sdof	15	0.831	13%	0.389	ridge: 0.627	c=0.36, β=1.05±0.14 (n=12)
forced/2dof	15	3.26	7%	6.25	ridge: 0.935	c=2.10, β=0.74±0.21 (n=10)

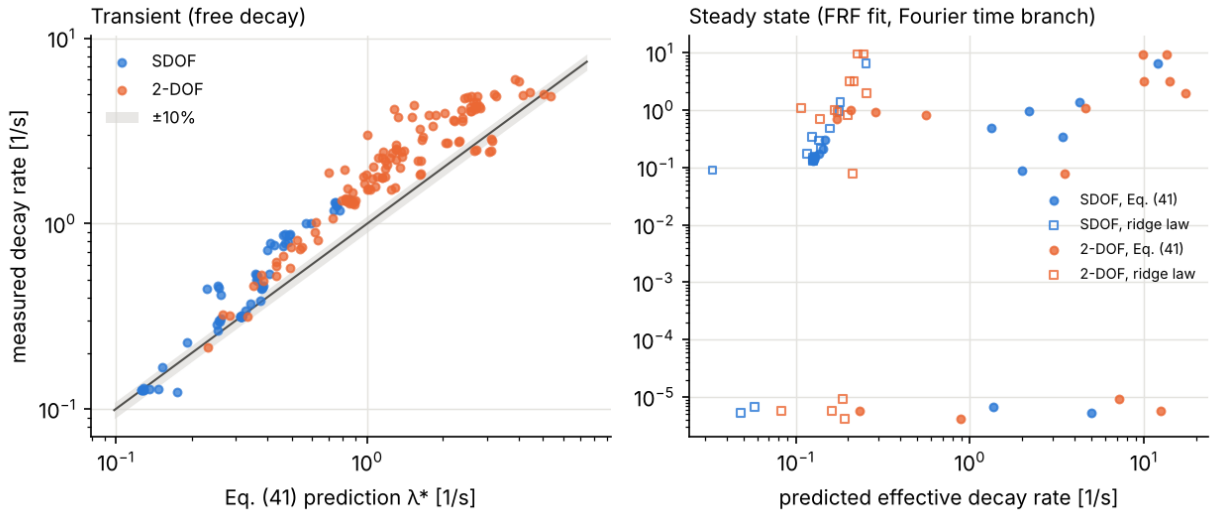
forced/sdof: point-wise ridge-law amplitude check on 66/135 non-collapsed points: median error 0.0398, 56% within 10%. Sweeps with no collapsed point: 6 (Eq. 41 median err 0.152, ridge 0.156).

forced/2dof: point-wise ridge-law amplitude check on 58/135 non-collapsed points: median error 0.697, 10% within 10%. Sweeps with no collapsed point: 5 (Eq. 41 median err 0.753, ridge 0.81).

Time-domain tanh-MLP PINN on forced SDOF at resonance (10 runs): amplitude ratio to exact 0.0021–0.0161, i.e. stuck at the trivial solution x = 0.

Exploratory breakdown (not pre-registered): transient runs binned by how well the PINN output is described by a single damped sinusoid, the shape Eq. (41)'s derivation assumes.

Setting	fit error bin	n	median $\lvert\Delta\rvert/\lambda$ (Eq. 41)	within 10%	median bias / λ_0	measured / predicted excess	c with β = 2
transient/sdof	[0.0,0.02)	21	6.248e-03	76%	6.752e-03	1.27	1.42
transient/sdof	[0.02,0.05)	10	0.151	30%	0.426	1.43	1.78
transient/sdof	[0.05,0.1)	9	0.329	0%	2.29	3.05	4.07
transient/sdof	[0.1,10.0)	20	0.412	0%	4.02	2.17	3.49
transient/2dof	[0.0,0.02)	2	0.0981	50%	1.13	1.04	–
transient/2dof	[0.02,0.05)	11	0.242	9%	1.37	1.58	1.9
transient/2dof	[0.05,0.1)	10	0.345	0%	4.1	1.86	2.58
transient/2dof	[0.1,10.0)	97	0.377	5%	5.7	1.82	2.85



Verdict. MET VIA FALLBACK: Eq. 41 misses 10%; measured relation reported; correction dropped (agreed 2026-10-11). Eq. (41) holds only where the bias is negligible (well-trained transient runs); wherever the bias matters, the measured excess decay rate is larger than Eq. (41) predicts from the measured residual floor, with roughly the same quadratic dependence on R but a coefficient above 1. In steady state neither Eq. (41) nor the ridge law predicts the FRF-fitted damping to 10%. Following the brief, no damping correction will be built on Eq. (41); later phases use the measured relation and direct measurement of the surrogate's effective damping.

5. (d) Literature check

No published stochastic or separable PINN identifying bearing or support parameters in rotating machinery or a railway wheelset was found (web search only; Crossref and publisher sites are blocked here). Closest: a deterministic PINN for a Jeffcott rotor (Parziale et al., Sensors 24:207, 2024) and gPC-Bayesian magnetic-bearing identification. The near-resonance PCE difficulty itself is established (Jacquelin et al., JSV 2015 and J. Eng. Mech. 2015/2016; Yaghoubi et al., PEM 2017; Kara and Ferguson, JSV 2023), so H1 is framed as a quantified diagnostic. Two full texts (J. Eng. Mech. 2016; PEM 2026) need checking with library access before the scaling law is claimed as new. Full table: docs/lit_review.md.

6. Provenance

Result file	git commit	wall time	timestamp (UTC)
results/p0/s2pinn_repro_seed42_nc4096.json	31c07e59cefb	5.22 h	2026-10-11T22:31:22Z
results/p0/s2pinn_cpu_timing.json	unknown	–	2026-10-11T13:21:12Z
results/p0/h1_sdof.json	unknown	0.19 h	2026-10-11T13:38:01Z
results/p0/h1_sdof_summary.json	unknown	–	2026-10-11T13:48:37Z
results/p0/h1_theory_check.json	848cee1563cc	–	2026-10-11T18:00:27Z
results/p0/eq41.json	a3ac030753f8…dirty	1.42 h	2026-10-11T15:13:35Z
results/gates/G0.json	31c07e59cefb	–	2026-10-11T22:31:41Z